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Section: BSIS – 2A

Reflection Essay:

The landscape of data management is shifting rapidly toward automation and infrastructure abstraction. Among the most transformative developments are Serverless Databases and AI-Integrated Databases. Serverless architecture allows developers to deploy databases without managing physical servers or provisioning fixed capacity, while AI integration embeds machine learning directly into the database engine to optimize performance and simplify user interactions. These trends represent a move away from manual maintenance toward a more responsive, intelligent data environment.

For our Inventory and Sales Management System, these technologies offer significant operational advantages. Implementing a serverless database, such as Aurora Serverless, would ensure the system automatically scales its resources during peak holiday sales or unexpected traffic spikes, maintaining performance without overpaying for idle capacity during slow periods. Simultaneously, AI-integrated features could revolutionize our analytics. Instead of writing complex SQL for every report, the system could utilize natural language processing to let managers ask, "Which products are likely to stock out next week?" by using predictive query tuning to analyze historical sales patterns and forecast reorder points with high precision.

Despite these benefits, several limitations must be considered. While serverless databases offer a "pay-as-you-go" model, costs can become unpredictable if traffic grows exponentially or if queries are inefficiently written. Furthermore, AI-integrated systems introduce a layer of complexity and a steep learning curve for the development team, requiring a deeper understanding of how the underlying models interpret data. There is also a significant risk of vendor lock-in; many of these advanced features are proprietary to specific cloud providers, making it difficult and expensive to migrate our system to another platform in the future if pricing or service quality changes.